

Artificial Intelligence & Machine Learning

Sample Knowledge Document for Testing a RAG Chatbot

This document contains short, factual notes designed for retrieval-augmented generation (RAG) testing. A chatbot can retrieve relevant passages from this PDF and use them as context when answering questions.

1. Artificial Intelligence

Artificial Intelligence (AI) is a field of computer science concerned with building systems that can perform tasks that normally require aspects of human intelligence. Examples include understanding language, recognizing images, planning, reasoning, and making decisions. AI is a broad field that includes machine learning, natural language processing, computer vision, robotics, and other areas.

2. Machine Learning

Machine Learning (ML) is a subfield of AI in which computer systems learn patterns from data instead of relying only on manually written rules. A machine learning model is trained using data and can then make predictions or decisions on new data. Common applications include spam detection, recommendation systems, fraud detection, forecasting, and image classification.

3. Main Types of Machine Learning

Supervised learning: The training data contains input examples together with known target labels or values. Classification and regression are common supervised-learning tasks.

Unsupervised learning: The data does not contain target labels. The system attempts to discover useful structure or patterns. Clustering is a common example.

Reinforcement learning: An agent interacts with an environment and learns from rewards or penalties. The goal is to learn a strategy that maximizes cumulative reward.

4. Neural Networks and Deep Learning

An artificial neural network is a computational model composed of connected units commonly called neurons. Neural networks usually contain an input layer, one or more hidden layers, and an output layer. During training, model parameters such as weights are adjusted to reduce a loss function.

Deep learning refers to machine learning methods based on neural networks with multiple layers. Deep learning has been especially successful in computer vision, speech processing, and natural language processing. Training large deep-learning models can require substantial amounts of data and computational power.

5. Natural Language Processing

Natural Language Processing (NLP) focuses on enabling computers to process and generate human language. NLP applications include chatbots, translation, text classification, summarization, information extraction, question answering, and sentiment analysis.

6. Embeddings and Vector Databases

An embedding is a numerical vector that represents information such as text in a mathematical space. Texts with similar meanings are intended to have embeddings that are relatively close according to a similarity measure. Embedding models are different from chat language models: an embedding model converts content into vectors, while a chat model generates or processes natural-language responses.

A vector database stores and searches vector representations. In a RAG system, document chunks can be converted into embeddings and stored in a vector database such as Chroma. When a user asks a question, the question is also embedded and the system retrieves document chunks that are similar to it.

7. Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation combines information retrieval with a generative language model. A basic RAG pipeline first loads documents, divides them into smaller chunks, creates embeddings for those chunks, and stores the embeddings in a vector store. At question time, a retriever selects relevant chunks. Those chunks are placed into the prompt as context, and a language model uses the context to produce an answer.

RAG can help a chatbot answer questions using a specific collection of documents instead of relying only on information learned during model training. The quality of a RAG system depends on factors such as document quality, chunking strategy, embedding quality, retrieval settings, and prompting.

8. Example RAG Components

Component	Purpose
PyPDFLoader	Loads text and metadata from PDF documents.
RecursiveCharacterTextSplitter	Divides document text into smaller overlapping chunks.
HuggingFaceEmbeddings	Creates numerical vector representations of text.
Chroma	Stores embeddings and supports similarity-based retrieval.
Retriever	Selects document chunks relevant to a user's question.
Groq chat model	Uses retrieved context to generate the final natural-language answer.

9. Test Questions

You can test a RAG chatbot with questions such as: What is machine learning? What are the three main types of machine learning? What is the difference between an embedding model and a chat model? What does Chroma do in a RAG system? How does a basic RAG pipeline work? What is NLP used for?

End of sample document.